

Zephylin Dusengimana

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RESEARCH INTERESTS

- Artificial Intelligence and Machine Learning, AI Security, Edge Intelligence, Large Language Models (LLMs), Cyber-Physical Systems

EDUCATION

Oklahoma Christian University <i>Master of Science in Computer Science GPA: 4.0/4.0</i>	Oklahoma, USA <i>Aug. 2024 – Dec. 2025</i>
Kyungdong University <i>Bachelor of Science in Computer Engineering GPA: 4.0/4.0</i>	Goseong, South Korea <i>Sep. 2020 – Aug. 2024</i>
Relevant Coursework: Cryptography & Cyber Security, Digital Forensics, Ethical Hacking, Computer Architecture & Digital Systems, Wireless Networks & Ubiquitous Computing, Internet of Things (IoT), Parallel & Distributed Computing, Artificial Intelligence & Deep Learning, Network Infrastructure & Management	

RESEARCH EXPERIENCE

Undergraduate Research Assistant <i>AI and Computer Vision Lab</i>	Mar. 2022 – May 2024 <i>Goseong, South Korea</i>
- Designed and evaluated a multi-scale object detection model (YOLO-based), improving detection accuracy by 25% across diverse environmental conditions - Conducted experimental analysis on large-scale structured and unstructured datasets to evaluate model robustness and generalization under adversarial and variable input conditions - Investigated model limitations and failure modes, contributing to iterative improvements through data-driven experimentation - Supported development of real-world AI systems under varying environmental constraints, analyzing deployment challenges	

PROFESSIONAL EXPERIENCE

AI Software Engineer <i>AfiaSync</i>	Jan. 2026 – Present <i>Atlanta, GA</i>
- Built an end-to-end clinical documentation pipeline using Azure OpenAI (GPT-4o) and Azure Speech Services, reducing practitioner documentation time from 15 min to 2 min per patient encounter - Engineered a versioned prompt framework with JSON schema enforcement and hallucination guardrails, increasing SOAP note quality scores from 40% to 80% on a standardized clinical rubric - Designed a Python LLM-as-a-judge evaluation pipeline benchmarking AI outputs across 30 clinical scenarios, cutting prompt iteration cycle time from hours to under 60 seconds	
AI Software Engineer Intern <i>Afia Sync</i>	June 2025 – Dec. 2025 <i>Atlanta, GA</i>
- Designed and implemented an AI-powered virtual health assistant using LLMs, improving response accuracy through retrieval-augmented generation (RAG) - Contributed to backend services development using .NET 8 and Azure Cosmos DB with multi-tenant data isolation, enforcing HIPAA-compliant partition-key segregation - Integrated AI-driven documentation workflows, reducing manual data entry and improving system efficiency - Automated end-to-end testing using Playwright, reducing manual testing workload by 95%	
Machine Learning Engineer Intern <i>HighView Tech Group</i>	Jan 2024 – Apr. 2024 <i>Kigali, Rwanda</i>
- Engineered a real-time anomaly detection system for fuel monitoring using Python and SQL - Built and optimized data pipelines for large-scale sensor data processing in IoT environments - Applied statistical and machine learning techniques to improve data quality and system reliability - Collaborated on system integration and performance optimization	

PROJECTS

Automated Attendance Management System | *Python, InsightFace, Redis, Streamlit*

- Designed an end-to-end face recognition pipeline using deep learning embeddings for real-time identification across 5,000+ records
- Implemented efficient data storage and retrieval using Redis for low-latency performance
- Achieved 96% recognition accuracy with robust validation and preprocessing techniques

RAG Chatbot | *Python, Streamlit, NLP, Hugging Face, LangChain, FAISS*

- Developed a multimodal RAG-based AI assistant integrating multiple LLM providers (OpenRouter, Groq) with a resilient fallback architecture for reliable real-time inference
- Engineered a document understanding pipeline for PDFs, images, and presentations using semantic retrieval, embeddings, and prompt engineering to improve question-answering accuracy
- Implemented AI system features including web search, conversational memory, speech interfaces, and automated document generation, emphasizing scalable human-centered AI design

PUBLICATIONS & RESEARCH WORK

- **Multi-Scale Cigarette Butt Detection Using YOLOv11: Bridging Scale and Domain Gaps** Under Preparation
- **Automatic Attendance System using Face Recognition (InsightFace-based)** Under Preparation

TECHNICAL SKILLS

Languages: Python, Java, C++, Prolog, SQL
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, RAG, OpenCV
Systems & Backend: FastAPI, Django, Spring Boot, REST APIs, Distributed Systems
Security & Networking: Cryptography, Network Security, Digital Forensics, IoT Protocols
Cloud & DevOps: AWS, Azure, Docker, CI/CD, Git
Databases: PostgreSQL, MySQL, MongoDB, Redis, Azure Cosmos DB

HONORS & AWARDS

- **Top Student**, Computer Engineering, Kyungdong University 2024
- **3× Winner**, KDU Hackathon Programming Contest, Kyungdong University 2021–2023
- **Winner**, RUDN International Math Olympiad, Russia 2020
- **Winner**, Rwanda National Mathematics Competition 2019